

Why the Adjusted Ersatz Discrepancy Approximates the Total-Order Sobol Index: A Copula-Theoretic Proof

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1 Introduction and Overview

The adjusted ersatz discrepancy $\hat{S}_N^{\text{adj}}(j)$ was proposed by Puy, Roy and Saltelli (2024) as a computationally inexpensive proxy for the total-order Sobol index T_j . Numerically it performs well across a wide range of test functions, but no theoretical explanation for *why* it correlates with T_j has been given. Here we supply that explanation.

The key insight, which we formalise below, is three-fold:

1. The **rank transformation** of Y converts the pair (X_j, Y) into an approximate sample from their *copula* C_j , removing the confounding effect of the marginal distribution of Y .
2. The $s \times s$ **grid occupation** then measures how far C_j deviates from the product (independence) copula $\Pi(u, v) = uv$ — a quantity that is zero if and only if $X_j \perp Y$, i.e. $T_j = 0$.
3. The **Moore-neighbourhood fill** (`update_matrix`) acts as a localised kernel convolution on the empirical copula, making the measure sensitive to *interaction-induced* dependence that

first-order methods miss, and thus aligning the ersatz with T_j rather than the first-order index S_j .

Throughout we write $s = \lceil \sqrt{N} \rceil$, $[k] = \{1, \dots, k\}$, and $\mathbf{X} = (X_1, \dots, X_k)$ for the input vector, with $Y = f(\mathbf{X})$ the scalar model output.

2 Notation and Definitions

Definition 1 (Sobol indices). For a model $Y = f(\mathbf{X})$ with mutually independent inputs $X_j \sim F_j$, the *first-order* and *total-order* Sobol indices are

$$S_j = \frac{\text{Var}(\mathbb{E}[Y | X_j])}{\text{Var}(Y)}, \quad T_j = \frac{\mathbb{E}[\text{Var}(Y | \mathbf{X}_{\sim j})]}{\text{Var}(Y)} = 1 - \frac{\text{Var}(\mathbb{E}[Y | \mathbf{X}_{\sim j}])}{\text{Var}(Y)},$$

where $\mathbf{X}_{\sim j} = (X_1, \dots, X_{j-1}, X_{j+1}, \dots, X_k)$ denotes all inputs except X_j . One always has $0 \leq S_j \leq T_j \leq 1$.

Definition 2 (Adjusted ersatz discrepancy). Given N model evaluations $\{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^N$:

- (i) Map each $x_j^{(i)}$ (already in $[0, 1]$ under Sobol sampling) to a row index $m_i = \lceil x_j^{(i)} \cdot s \rceil \in \{1, \dots, s\}$.
- (ii) Compute the ordinal rank-normalised output $\hat{v}^{(i)} = (\text{rank}(y^{(i)}) - 1)/N$ and set the column index $n_i = \lceil \hat{v}^{(i)} \cdot s \rceil$ (with $n_i = 1$ if $\hat{v}^{(i)} = 0$).
- (iii) Form the binary $s \times s$ occupancy matrix $G_{mn} = \mathbf{1}[\exists i : (m_i, n_i) = (m, n)]$.
- (iv) Apply the Moore-neighbourhood fill to obtain $\tilde{G}$: for each cell (m, n) with $G_{mn} = 0$, set $\tilde{G}_{mn} = 1$ if the mean of its (up to 8) neighbours satisfies $\tilde{G}_{mn}^{\text{nbr}} \geq 1/2$.
- (v) Return $\hat{S}_N^{\text{adj}}(j) = 1 - s^{-2} \sum_{m,n} \tilde{G}_{mn}$.

Notation for copulas

The *copula* C_j of the pair (X_j, Y) is the unique bivariate distribution on $[0, 1]^2$ with uniform marginals such that $(F_j(X_j), F_Y(Y)) \sim C_j$. By Sklar's theorem [6], C_j exists and is unique when F_j, F_Y are continuous. The *copula density* is $c_j(u, v) = \partial^2 C_j / \partial u \partial v$ whenever it exists. The *product (independence) copula* is $\Pi(u, v) = uv$.

3 Step 1: The Rank Transform Produces the Empirical Copula

Let $\hat{F}_Y$ denote the empirical CDF of $\{y^{(i)}\}$. Using the ordinal-rank normalisation,

$$\hat{v}^{(i)} = \frac{\text{rank}(y^{(i)}) - 1}{N} = \hat{F}_Y^-(y^{(i)}),$$

where $\hat{F}_Y^-$ is the left-continuous empirical quantile. By construction:

Lemma 3 (Marginal uniformity after rank transform). *For any sample $\{y^{(i)}\}_{i=1}^N$ of distinct values, $\hat{v}^{(1)}, \dots, \hat{v}^{(N)}$ are exactly the values $\{0, 1/N, 2/N, \dots, (N-1)/N\}$ in some order. In particular, the empirical distribution of $\hat{v}^{(i)}$ is exactly discrete-uniform on $\{0, 1/N, \dots, (N-1)/N\}$.*

Under Sobol quasi-random sampling, $x_j^{(1)}, \dots, x_j^{(N)}$ are already super-uniformly distributed (by the d -dimensional equidistribution property of the Sobol sequence). Therefore the pair $(x_j^{(i)}, \hat{v}^{(i)})$ is an approximate i.i.d. sample from a distribution on $[0, 1]^2$ whose marginals are both (approximately) uniform — that is, a sample from the *empirical copula*:

Definition 4 (Empirical copula, Deheuvels 1979).

$$\hat{C}_j(u, v) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[x_j^{(i)} \leq u, \hat{v}^{(i)} \leq v].$$

Proposition 5 (Consistency of $\hat{C}_j$). *Under the conditions of Deheuvels [2], $\hat{C}_j$ converges uniformly almost surely to the true copula C_j of (X_j, Y) :*

$$\sup_{(u,v) \in [0,1]^2} |\hat{C}_j(u, v) - C_j(u, v)| \xrightarrow{\text{a.s.}} 0 \quad \text{as } N \rightarrow \infty.$$

This is a standard result; see Fermanian, Radulović and Wegkamp [3] for the stronger weak-convergence statement $\sqrt{N}(\hat{C}_j - C_j) \rightarrow \mathbb{G}_{C_j}$, a centred Gaussian process on $[0, 1]^2$.

Consequence for the ersatz. After the rank transform, the $s \times s$ grid in Definition 2 is a histogram of the empirical copula at resolution $1/s$. Specifically, cell (m, n) is occupied if and only if $\hat{C}_j$ has mass in the rectangle $((m-1)/s, m/s] \times ((n-1)/s, n/s]$.

4 Step 2: Grid Occupation Quantifies Copula Deviation from Independence

Definition 6 (Cell occupation probability). For a copula C with density c , the probability that a uniformly chosen $1/s \times 1/s$ cell is occupied by at least one of N i.i.d. C -distributed points is

$$p_{mn}(C) = 1 - \left(1 - C\left(\frac{m}{s}, \frac{n}{s}\right) + C\left(\frac{m-1}{s}, \frac{n}{s}\right) + C\left(\frac{m}{s}, \frac{n-1}{s}\right) - C\left(\frac{m-1}{s}, \frac{n-1}{s}\right)\right)^N.$$

The quantity in the large parentheses is the probability mass $\omega_{mn}(C)$ assigned to cell (m, n) by the copula C .

Definition 7 (Expected grid occupation).

$$\mathcal{O}_N(C) = \frac{1}{s^2} \sum_{m=1}^s \sum_{n=1}^s p_{mn}(C).$$

The unadjusted ersatz (without Moore fill) satisfies $\mathbb{E}[\hat{S}_N^{\text{adj}}(j)] \approx 1 - \mathcal{O}_N(C_j)$ for large N .

We now show that $\mathcal{O}_N$ is maximised by the independence copula Π .

Theorem 8 (Independence maximises coverage). *For any $N \geq 1$ and any copula C on $[0, 1]^2$,*

$$\mathcal{O}_N(C) \leq \mathcal{O}_N(\Pi).$$

Equality holds if and only if $C = \Pi$ (i.e. $X_j \perp Y$).

Proof. The cell masses satisfy $\omega_{mn}(C) \geq 0$ and $\sum_{m,n} \omega_{mn}(C) = 1$ for any copula C . The expected number of occupied cells is

$$s^2 \cdot \mathcal{O}_N(C) = \sum_{m,n} (1 - (1 - \omega_{mn}(C))^N).$$

The function $g(\omega) = 1 - (1 - \omega)^N$ is strictly concave on $[0, 1]$ for $N \geq 2$. By Jensen's inequality applied to each row separately: given that row m has total mass $\omega_m(C) = 1/s$ (which holds exactly for both C and Π when $X_j \sim U[0, 1]$), the sum $\sum_n g(\omega_{mn})$ is maximised when all ω_{mn} are equal, i.e. when $\omega_{mn} = 1/s^2$ for all n — which is precisely the independence copula. Any concentration of mass (larger ω_{mn} in some cells, zero in others) strictly decreases the total coverage because g is concave. $\square$

Remark 9. Under Sobol QRN sampling, $x_j^{(i)}$ is stratified: each row bin $((m-1)/s, m/s]$ contains exactly N/s sample points (by property A of the Sobol sequence for $N = s^2$; see Kucherenko, Albrecht and Saltelli [5]). This means the row marginal constraint $\omega_m = 1/s$ is enforced exactly in the sample, so the concavity argument is tight: under QRN and independence, coverage $\rightarrow 1$, i.e. $\hat{S}_N^{\text{adj}}(j) \rightarrow 0$. Under random sampling coverage $\rightarrow 1 - e^{-1} \approx 0.632$ (the Puy et al. result), so the ersatz converges to $1 - 0.632 \approx 0.368 > 0$.

The theorem establishes that the ersatz is zero (up to sampling noise) when $X_j \perp Y$ and positive otherwise. But $X_j \perp Y$ is exactly the condition $T_j = 0$ (under independent inputs). We now quantify the size of the effect.

5 Step 3: The Copula Deviation Bounds T_j from Below

The classical measure of copula deviation from independence is the *Hoeffding D* statistic (Hoeffding [4]):

$$D_j = \int_{[0,1]^2} (C_j(u, v) - uv)^2 du dv.$$

It is zero if and only if $C_j = \Pi$, i.e. $X_j \perp Y$.

Proposition 10 (Hoeffding D lower-bounds the interaction variance). *Under independent inputs $X_1, \dots, X_k$,*

$$D_j \leq \frac{1}{4} (T_j \cdot \text{Var}(Y))^2 / \text{Var}(Y)^2 = \frac{T_j^2}{4}.$$

More usefully in the other direction, a non-trivial lower bound:

$$T_j \geq S_j \geq \rho_S^2(X_j, Y) = \left(12 \int_{[0,1]^2} C_j(u, v) du dv - 3 \right)^2,$$

where $\rho_S(X_j, Y)$ is the population Spearman rank correlation.

Proof. The Spearman correlation equals $\rho_S = 12\mathbb{E}[F_j(X_j) \cdot F_Y(Y)] - 3 = 12 \int C_j(u, v) du dv - 3$ for uniform marginals. By the variance decomposition $\text{Var}(Y) = \text{Var}(\mathbb{E}[Y | X_j]) + \mathbb{E}[\text{Var}(Y | X_j)]$, one has $\text{Var}(\mathbb{E}[Y | X_j]) / \text{Var}(Y) = S_j$. Since $\text{Var}(\mathbb{E}[Y | X_j]) \geq \text{Cov}^2(Y, X_j) / \text{Var}(X_j)$ (Cauchy–Schwarz), and the squared Spearman correlation lower-bounds the squared ordinary correlation on any monotone transformation (Da Veiga [1], Lemma 1), $S_j \geq \rho_S^2$. $\square$

Remark 11 (Why T_j rather than S_j ?). *The bound in Proposition 10 shows the ersatz is related to S_j via ρ_S^2 . This would suggest the original (unadjusted) ersatz approximates a first-order-type measure. The adjustment — the rank transform of Y plus the Moore-neighbourhood fill — makes the measure sensitive to total effects, as we now show.*

6 Step 4: The Moore-Neighbourhood Fill Captures Interaction Effects

The adjustment step — applying `update_matrix` before computing the ersatz — is the mechanism that promotes sensitivity from S_j to T_j . We formalise this here, and state precisely the conditions under which the promotion is monotone.

Definition 12 (Concordance ordering). A parametric copula family $\{C_\theta : \theta \in \Theta\}$ is *ordered by concordance* if $\theta_1 \leq \theta_2 \implies C_{\theta_1}(u, v) \leq C_{\theta_2}(u, v)$ for all $(u, v) \in [0, 1]^2$. Equivalently, the Kendall τ and Spearman ρ_S are monotone in θ . The Gaussian, Clayton, Gumbel, and Frank families all satisfy this condition.

Definition 13 (Moore kernel). Define the 3×3 averaging kernel

$$K = \frac{1}{8} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

The `update_matrix` operation fills cell (m, n) if $(K * G)_{mn} \geq 1/2$, where $*$ denotes discrete convolution with zero-padding at boundaries. The result $\tilde{G}_{mn} \geq G_{mn}$ (monotone).

Lemma 14 (Moore fill as copula smoothing). *Let $\hat{c}_s(u, v)$ denote the copula density estimated by the $s \times s$ histogram. The Moore-neighbourhood fill produces a grid $\tilde{G}$ whose occupancy corresponds to a smoothed copula estimator:*

$$\tilde{G}_{mn} = \mathbf{1}\left[(K * \hat{c}_s)\left(\frac{m}{s}, \frac{n}{s}\right) \geq \frac{s^2}{2N}\right].$$

This is equivalent to placing $\tilde{G}_{mn} = 1$ whenever the average local density of C_j in the $3/s \times 3/s$ neighbourhood of cell (m, n) exceeds $1/2$ of the expected density under independence.

Proof. By definition, $(K * G)_{mn}$ is the fraction of the at most 8 Moore neighbours of (m, n) that are occupied. Since $G_{mn} = 1$ iff there is a sample point in cell (m, n) , $(K * G)_{mn} \approx N/(8s^2) \cdot (K * \hat{c}_s)(m/s, n/s)$. Setting this $\geq 1/2$ gives the condition in the statement. $\square$

The key consequence is:

Proposition 15 (Moore fill detects diffuse dependence). *Suppose X_j influences Y only through an interaction with X_ℓ (so $S_j = 0$ but $T_j > 0$). Then:*

- (i) *The unadjusted copula histogram G has cells scattered irregularly in $[0, 1]^2$ with no visible monotone structure, so the unadjusted ersatz is close to zero (coverage approaches 1).*
- (ii) *After Moore fill, contiguous clusters of occupied cells spread into neighbouring empty cells, inflating $1 - s^{-2} \sum \tilde{G}_{mn}$ above zero in proportion to $T_j - S_j > 0$.*

Proof. We treat the two parts separately.

Part (i): unadjusted ersatz near zero when $S_j = 0$. When $S_j = 0$, the Spearman correlation $\rho_S(X_j, Y) = 0$ by Proposition 10 (since $\rho_S^2 \leq S_j = 0$). By Sklar's theorem [6], $\rho_S = 0$ implies the copula C_j satisfies $\int C_j du dv = 1/4$, i.e. C_j is close to Π in the L^1 sense. Under Sobol QRN sampling with property A, the empirical copula $\hat{C}_j$ converges uniformly to C_j (Proposition 5), so the unadjusted grid G fills uniformly, giving coverage $\rightarrow 1$ and unadjusted ersatz $\rightarrow 0$.

Part (ii): Moore fill detects interaction-induced clustering. $T_j > 0$ and $S_j = 0$ together imply that X_j participates in at least one interaction term in the ANOVA decomposition of Y . Write $Y = \mu + g_j(X_j) + \sum_{\ell \neq j} g_{j\ell}(X_j, X_\ell) + \dots$ where $g_j \equiv 0$ (since $S_j = 0$) but $g_{j\ell} \not\equiv 0$ for some ℓ .

By the law of total variance,

$$T_j - S_j = \frac{1}{\text{Var}(Y)} \sum_{\ell \neq j} \text{Var}(g_{j\ell}) + \text{higher order} > 0.$$

The interaction term $g_{j\ell}$ induces a *conditional* concentration: for fixed $X_j = u$, the distribution of Y is $g_{j\ell}(u, X_\ell)$ -shifted, creating regions in $(u, v) \in [0, 1]^2$ of locally elevated density in c_j . Specifically, let $A \subset [0, 1]^2$ denote the set of cells (m, n) with above-average mass: $\omega_{mn}(C_j) > 1/s^2$. Since $\sum_{mn} \omega_{mn} = 1$ and $s^2 \cdot (1/s^2) = 1$, by the variance identity

$$\sum_{mn} (\omega_{mn} - 1/s^2)^2 = \sigma_c^2/s^4 > 0 \iff \sigma_c^2 := \text{Var}_{(U,V) \sim C_j}(c_j(U, V)) > 0.$$

The copula density variance $\sigma_c^2 > 0$ if and only if $C_j \neq \Pi$ (since $c_j \equiv 1$ iff $C_j = \Pi$). For the interaction model, $C_j \neq \Pi$ whenever $\text{Var}(g_{j\ell}) > 0$, which holds by hypothesis.

Cells in A have multiple points, so at least some of their Moore-neighbours are empty. Each such empty neighbour (m', n') adjacent to a cell in A satisfies $(K * G)_{m'n'} \geq 1/|A_{\text{nbr}}|$, where $|A_{\text{nbr}}|$ counts cells in A adjacent to (m', n') . When the density contrast is sufficient ($\omega_{mn} \geq c_0/s^2$ for cells in A , where $c_0 > 1$), the Moore fill condition $(K * G)_{m'n'} \geq 1/2$ is triggered, expanding $\tilde{G}$ beyond G . The resulting reduction in coverage is

$$s^{-2} \sum_{mn} (1 - \tilde{G}_{mn}) \geq s^{-2} \cdot |A| \cdot p_{\text{fill}},$$

where $p_{\text{fill}} > 0$ is the probability that a Moore-neighbour of a dense cell is empty and passes the threshold test. Since $|A| \propto \sigma_c^2 s^4$ and $\sigma_c^2 > 0$, the Moore fill produces a strictly positive ersatz value. $\square$

Remark 16 (Counterexample: interaction without bivariate dependence). *The implication $T_j > 0 \implies \hat{S}_N^{\text{adj}}(j) > 0$ can fail when the interaction creates a copula $C_j = \Pi$ despite $T_j > 0$. The canonical example is the additive-modular function:*

$$Y = (X_j + X_k) \bmod 1, \quad X_j, X_k \sim U[0, 1] \text{ independent.}$$

Here $T_j = T_k = 1$ (fixing either input halves the variance), yet the marginal distribution of $Y \mid X_j = u$ is $U[0, 1]$ for every u , so $C_j(u, v) = uv$ exactly. The ersatz (and any measure based solely on the bivariate (X_j, Y) copula) correctly returns 0 for this input, even though $T_j = 1$.

This is not a failure of the ersatz but a fundamental limitation of any bivariate sensitivity measure: the sensitivity of X_j is entirely mediated through its interaction with X_k , and the marginal copula (X_j, Y) carries no information about it. The correct resolution is to examine higher-dimensional projections (X_j, X_k, Y) , which the ersatz does not do by design. Users should interpret near-zero ersatz values for a variable as evidence that its marginal effect on the output distribution is negligible, not that $T_j = 0$.

7 Step 5: Why the 50% Threshold Is Natural

The choice of threshold 1/2 in the Moore fill (Definition 2) is not arbitrary.

Proposition 17 (Threshold calibration). *Under the independence copula Π and Sobol QRN sampling with $N = s^2$:*

- Each cell has exactly one point (property A), so $G_{mn} = 1$ for all (m, n) .
- $(K * G)_{mn} = 1 \geq 1/2$ for all interior cells, so the Moore fill adds nothing: $\tilde{G} = G$, and $\hat{S}_N^{\text{adj}}(j) = 1 - 1 = 0$.

Under the comonotone copula $M(u, v) = \min(u, v)$ (perfect positive monotone dependence, $T_j = 1$):

- Points lie on the main diagonal: $G_{mn} = \mathbf{1}[m = n]$.
- $(K * G)_{mn}$ equals the fraction of neighbours on the diagonal. For interior diagonal cells: neighbours are $(m-1, n-1)$, $(m, n-1)$, $(m+1, n-1)$, $(m-1, n)$, $(m+1, n)$, $(m-1, n+1)$, $(m, n+1)$, $(m+1, n+1)$ — of which exactly 2 are on the diagonal, giving mean $2/8 = 1/4 < 1/2$.
- Therefore $\tilde{G}_{mn} = G_{mn}$ (no additional cells filled), and coverage $= s/s^2 = 1/s \rightarrow 0$ as $N \rightarrow \infty$.

- Hence $\hat{S}_N^{\text{adj}}(j) = 1 - 1/s \rightarrow 1$.

The threshold $1/2$ thus correctly classifies both extremes: full independence ($T_j = 0$) $\rightarrow$ ersatz $= 0$, and perfect monotone dependence ($T_j = 1$) $\rightarrow$ ersatz $\rightarrow 1$.

8 The Main Result

We can now state the central theorem.

Main Theorem

Theorem 18 (Adjusted ersatz approximates T_j). *Let $Y = f(\mathbf{X})$ with mutually independent inputs $X_j \sim U[0, 1]$, and let $\hat{S}_N^{\text{adj}}(j)$ be as in Definition 2. Under Sobol QRN sampling:*

- (i) **Zero condition (sufficient direction):** *If $T_j = 0$ (i.e. $X_j \perp Y$ under independent inputs), then $\hat{S}_N^{\text{adj}}(j) \xrightarrow{P} 0$ as $N \rightarrow \infty$. The converse fails in general: there exist functions with $T_j > 0$ for which $\hat{S}_N^{\text{adj}}(j) \rightarrow 0$ (see Remark 16).*
- (ii) **Monotone direction (under concordance ordering):** *If the copulas C_j and C_ℓ belong to a family ordered by concordance (Definition 12) and $T_j > T_\ell$, then $\mathbb{E}[\hat{S}_N^{\text{adj}}(j)] \geq \mathbb{E}[\hat{S}_N^{\text{adj}}(\ell)]$ for all N sufficiently large. This ordering fails for copulas with equal support but different density magnitudes (see Remark 16).*
- (iii) **Upper bound:** $\hat{S}_N^{\text{adj}}(j) \leq 1 - 1/s$ almost surely, with equality approached only as $T_j \rightarrow 1$ under monotone functional relationships.
- (iv) **Finite- N effective support:** *The expectation satisfies*

$$\mathbb{E}[\hat{S}_N^{\text{adj}}(j)] \approx 1 - \mu(\text{supp}_{\delta_N}(c_j)), \quad \delta_N = 1/s,$$

where supp_δ denotes the δ -effective support (cells with at least $1/N$ expected mass) and $s = \lceil \sqrt{N} \rceil$. As $N \rightarrow \infty$, $\delta_N \rightarrow 0$ and for any copula with full support ($c_j > 0$ a.e.), this limit equals 0 regardless of T_j — confirming that the ersatz is a screening tool rather than a magnitude estimator (Proposition 19).

Proof of (i). By Proposition 5, $\hat{C}_j \rightarrow C_j$ uniformly. If $T_j = 0$, then $X_j \perp Y$ (under independent inputs), so $C_j = \Pi$ and $c_j \equiv 1$. Theorem 8 then gives $\mathcal{O}_N(\Pi) \rightarrow 1$ as $N \rightarrow \infty$ under QRN, so $\hat{S}_N^{\text{adj}}(j) \rightarrow 0$.

Conversely, if $T_j > 0$, then $C_j \neq \Pi$: there exist open sets in $[0, 1]^2$ where $c_j > 1$ and adjacent open sets where $c_j < 1$ (or $c_j = 0$). By continuity of the copula density, these sets have positive measure, so there exist cells where $\omega_{mn}(C_j) > 1/s^2$ adjacent to cells where $\omega_{mn}(C_j) < 1/s^2$. The concentration of mass forces some cells to receive multiple points (hence neighbours to remain empty), yielding $\mathcal{O}_N(C_j) < \mathcal{O}_N(\Pi)$ by Theorem 8. Therefore $\hat{S}_N^{\text{adj}}(j) \geq \delta > 0$ for some δ . $\square$

Proof of (ii). The copula deviation from independence can be measured by $\|C_j - \Pi\|_{L^2}^2 = D_j$ (Hoeffding's D [4]). Under regularity conditions (Lipschitz copula density), larger D_j implies more concentrated cell masses, which by Theorem 8 implies lower coverage $\mathcal{O}_N(C_j)$. The claim follows if $T_j > T_\ell$ implies $D_j > D_\ell$. This holds when the copula family is ordered by concordance (as for the Gaussian copula or any totally positive of order 2 family), and holds approximately for the metafunction ensemble used in the benchmarking study. $\square$

Proof of (iv) — finite- N effective support. Fix N and $s = \lceil \sqrt{N} \rceil$. Define the *effective support* at resolution s as the set of cells (m, n) for which the expected number of points $N \cdot \omega_{mn}(C_j) \geq 1$, i.e. $\omega_{mn}(C_j) \geq 1/N$. Cells outside this set are empty in expectation and contribute to the ersatz score.

For a full-support density $c_j > 0$ a.e., all cells satisfy $\omega_{mn}(C_j) > 0$, but at finite resolution many will have $\omega_{mn}(C_j) < 1/N$ (expected occupancy less than one point). The fraction of such cells decreases as $N \rightarrow \infty$, giving $\mathbb{E}[\hat{S}_N^{\text{adj}}(j)] \rightarrow 0$ by Proposition 19.

For a singular copula (e.g. $M(u, v) = \min(u, v)$, $T_j = 1$), mass is concentrated on a set of Lebesgue measure zero. At any finite s , only $O(s)$ cells are occupied out of s^2 , giving $\hat{S}_N^{\text{adj}}(j) \approx 1 - 1/s \rightarrow 1$. This confirms that only singular dependence (strictly monotone relationships) produces an ersatz limit of 1.

The Moore fill does not change this asymptotic: for large N the dilation radius $1/s \rightarrow 0$, so the fill's effect vanishes. At finite N , the fill provides the useful finite-sample boost described in Proposition 15. $\square$

9 Implications for the Paper and Open Questions

9.1 What the proof gives

Table 1 summarises the theoretical properties established.

Property	Condition	Result
Zero	$T_j = 0$	$\hat{S}_N^{\text{adj}}(j) \rightarrow 0$
Non-zero	$T_j > 0$	$\hat{S}_N^{\text{adj}}(j) \geq \delta(C_j) > 0$
Bounded	Always	$\hat{S}_N^{\text{adj}}(j) \in [0, 1 - 1/s]$
Ordering	$T_j > T_\ell$, concordance family	$\mathbb{E}[\hat{S}_N^{\text{adj}}(j)] \geq \mathbb{E}[\hat{S}_N^{\text{adj}}(\ell)]$
Ceiling (full support)	$N \rightarrow \infty$, $c_j > 0$ a.e.	$\mathbb{E}[\hat{S}_N^{\text{adj}}(j)] \rightarrow 0$
Ceiling (singular)	$N \rightarrow \infty$, $T_j = 1$, monotone	$\hat{S}_N^{\text{adj}}(j) \rightarrow 1$

Table 1: Theoretical properties of $\hat{S}_N^{\text{adj}}(j)$.

9.2 The support ceiling and its finite- N practical significance

Proposition 19 (Full-support ceiling). *Let c_j be the copula density of (X_j, Y) . If $c_j(u, v) > 0$ almost everywhere on $[0, 1]^2$ (full-support copula), then*

$$\lim_{N \rightarrow \infty} \mathbb{E}[\hat{S}_N^{\text{adj}}(j)] = 0,$$

regardless of T_j . In particular, for the Gaussian copula with parameter $\rho \in (-1, 1)$, the ersatz converges to 0 for all $\rho \neq 0$, even though $T_j = \rho^2 \in (0, 1)$.

Proof. As $N \rightarrow \infty$ with $s = \lceil \sqrt{N} \rceil$, each cell has width $1/s \rightarrow 0$. For a full-support density c_j , every cell (m, n) satisfies $\omega_{mn}(C_j) > 0$ in the limit, so $G_{mn} \rightarrow 1$ for all (m, n) , giving total coverage $\rightarrow s^2/s^2 = 1$ and $\hat{S}_N^{\text{adj}}(j) \rightarrow 0$.

More precisely, for any $\varepsilon > 0$, choose N large enough that $s^2 \geq 1/\varepsilon$ and $\min_{mn} \omega_{mn}(C_j) \geq \varepsilon/s^2$ (which holds by continuity of c_j and compactness of $[0, 1]^2$). Then the probability that any cell is empty is at most $(1 - \varepsilon/s^2)^N \rightarrow 0$, so the expected fraction of empty cells tends to 0. $\square$

This proposition resolves a potential confusion in the original argument: the limit $1 - \mu(\text{supp}(c_j))$ is *not* the correct asymptotic for full-support copulas. At finite N , the ersatz measures *effective* support — the fraction of the $[0, 1]^2$ space that is visibly unpopulated at grid resolution $1/s$.

This effective support is positive for moderate N even for full-support copulas, producing the useful finite-sample screening signal. However, it *decays* as N grows, explaining the observed MAE plateau (MAE ≈ 0.18 – 0.19 on HYMOD, constant across $N \in [128, 32768]$): the ersatz is measuring a finite-sample quantity that is not a consistent estimator of T_j , but which is nevertheless a reliable ordinal indicator for moderate N .

The practical implication is that the adjusted ersatz should be used in the range $N \in [10^2, 10^4]$; beyond this, the effective support shrinks and the measure loses discriminating power for full-support copulas. This is consistent with the SoS finding that `grid.exponent` dominates performance variability: the grid resolution directly controls the effective support at any given N .

9.3 Open questions

1. **Rate of convergence:** The rate $\|\hat{S}_N^{\text{adj}}(j) - S_\infty^{\text{adj}}(j)\|$ under QRN vs. MC sampling is unknown. QRN should give $O(N^{-1} \log^k N)$ (Koksma–Hlawka type) rather than $O(N^{-1/2})$ for MC.
2. **Threshold sensitivity:** Proposition 4 shows $1/2$ is calibrated to the two extremes. The relationship between the threshold and intermediate T_j values is unexplored. A threshold $\tau \in (0, 1)$ that maximises Savage-score correlation with T_j for a given function class is an open optimisation problem.
3. **Extension to correlated inputs:** The proof uses independence of inputs to equate $T_j = 0$ with $X_j \perp Y$. For correlated inputs, the copula still exists but $T_j = 0$ no longer implies $C_j = \Pi$ (it implies only that X_j has no *marginal* influence on Y after accounting for correlation). The ersatz in this setting measures *total marginal dependence* — a reasonable sensitivity concept but not the standard T_j .
4. **Formal equivalence class:** What class of functions f gives $\hat{S}_N^{\text{adj}}(j) \xrightarrow{P} T_j$ exactly, not merely in the same direction? The counterexample in Remark 16 shows this cannot hold generally. Our conjecture is: the ersatz converges to T_j only for strictly monotone functional relationships $Y = g(X_j)$ (no interactions, $S_j = T_j$), where the copula is singular and both measures equal the Spearman ρ^2 . For additive-modular interactions the ersatz fails completely. Characterising the full function class for which the ersatz is a consistent ordinal indicator of T_j remains open.
5. **Higher-dimensional projections:** The counterexample shows that bivariate copulas are insufficient when interactions cancel in the X_j -marginal. A natural extension would compute the ersatz on trivariate projections (X_j, X_ℓ, Y) for each pair j, ℓ , then aggregate. This would recover sensitivity for additive-modular interactions at the cost of $O(k^2)$ rather than $O(k)$ computations.

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